

Nathan Chung

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EDUCATION

Texas A&M University

Bachelor of Science in Computer Science - GPA: 3.63/4.0

College Station, TX

Expected May 2029

SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C/C++, C#

Frameworks/Libraries: FastAPI, Streamlit, SQLAlchemy, Pydantic, Plotly, React, Node.js, Next.js, TailwindCSS

Developer Tools: Git, Docker, GitHub Actions, pytest, ruff, MongoDB, REST APIs, WebSocket

AI/ML: pandas, scikit-learn, Claude API, OpenAI API, ChromaDB

EXPERIENCE

AMD (Advanced Micro Devices, Inc.)

Jan 2026 – Present

Software Engineer Intern

Austin, TX

- Built Python feature-validation scripts to configure and validate CPU power-management features on remote dual-socket Linux servers, enabling repeatable regression testing across 10+ silicon features.
- Identified and resolved critical bugs across multiple codebases and caught 8 incorrect register references via code review, migrated a register access library to fix dual-socket data misidentification, and debugged a silent write failure caused by an inverted control bit.
- Built feature validation scripts from scratch and contributed directly to the team's shared GitHub repository, including adding dual-socket support for multi-IOD hardware configurations.
- Built AI-powered Python tools to auto-generate test configurations from specifications and validate feature scripts for correctness, reducing manual setup across a distributed team spanning US, Malaysia, and India.

Arkane Digital

Jun 2025 – Aug 2025

Software Engineer Intern

Remote

- Developed and deployed 10+ AI-powered workflows on the Raikoo platform using Python and scikit-learn to automate portions of the front-end design-to-code pipeline.
- Built multi-component generation logic with text-preprocessing pipelines for the design-to-code system, supporting 20+ UI configurations across internal product teams.

PROJECTS

TelemetryBench | *Python, FastAPI, Streamlit, Docker, SQLite, pandas, scikit-learn, GitHub Actions*

- Built a Python telemetry analysis tool (FastAPI + Streamlit + SQLAlchemy/SQLite) that ingests raw DAQ CSVs from hardware parameter sweeps and exposes 8 REST endpoints for sweep ingestion, comparison, time-series queries, and anomaly scoring.
- Added ML-based run-quality scoring (scikit-learn Isolation Forest over 4 per-metric statistics) exposed via REST and surfaced as a sortable dashboard column — catches drift cases like normal average but high variance that static thresholds miss.
- Containerized with Docker Compose (2-service stack, persistent SQLite volume) and gated every commit with a GitHub Actions CI pipeline running pytest, ruff lint, and Docker build.

AI Social Media Workflow | *OpenAI API, Gemini 2.5 Flash, TensorFlow*

- Developed an AI-driven workflow on Raikoo AI that automated cross-platform content generation (LinkedIn, Instagram, X, blogs, articles), reducing manual creation time by 70% and boosting content quality by 30% during internal testing.
- Integrated Gemini 2.5 Flash, GPT-4, and vision models to generate API-compliant captions, images, and JSON payloads, improving automation accuracy and deployment reliability by 40% through a built-in content review interface.